

Finite-dimensionality transfers across a sphere isometry

MathlibAnnex.Sphere.finiteDimensional_codomain

theorem

Uses the radial homeomorphism to transfer local compactness, then applies the finite-dimensionality criterion for real normed spaces.

Statement

Let X, Y be real normed spaces. Put $S_X = \{x \in X : \|x\|_X = 1\}$ and $S_Y = \{y \in Y : \|y\|_Y = 1\}$, each with its ambient chord distance. Let $\Delta : S_X \rightarrow S_Y$ be a bijective isometry of their unit spheres with the ambient chord metrics. If X is finite-dimensional over $\mathbb{R}$, then Y is finite-dimensional over $\mathbb{R}$.

Assumptions

Only X is assumed finite-dimensional at the start. Both spaces have real normed-vector-space structures, and Δ is a sphere isometry equivalence. There is no initial completeness or finite-dimensionality assumption on Y ; the zero-dimensional case is included. The two carriers may have different universes in the formal statement.

Conclusion

The codomain Y is finite-dimensional. Applying the theorem to Δ^{-1} transfers finite-dimensionality in the opposite direction; that is the separately cited domain theorem.

Notes

The map used to carry the topology is the radial homeomorphism R_Δ , with inverse $R_{\Delta^{-1}}$. This step asserts transfer of finite-dimensionality; a linear isometry is not constructed here.

Proof route

For clarity, the radial map is $R_\Delta(0) = 0$ and $R_\Delta(x) = \|x\|_X \Delta(x/\|x\|_X)$ for $x \neq 0$, with inverse $R_{\Delta^{-1}}$. Use the homeomorphism $R_\Delta : X \rightarrow Y$, whose continuity and inverse continuity follow from the two 3-Lipschitz estimates. Finite-dimensional real normed spaces are proper, so every closed bounded ball in X is compact. Each point therefore has a compact neighbourhood, making X locally compact. A homeomorphism carries a compact neighbourhood to a compact neighbourhood, so Y is locally compact. The local-compactness criterion for a real normed space now implies that Y is finite-dimensional:

$$\begin{aligned} \dim_{\mathbb{R}} X < \infty &\implies X \text{ locally compact} \\ \implies Y \text{ locally compact} &\implies \dim_{\mathbb{R}} Y < \infty. \end{aligned}$$

▼ **Proof steps**

- 1. Finite-dimensionality of X supplies its proper-space structure. This is where the dimension hypothesis is used.
- 2. Transfer local compactness through R_Δ . The criterion is then applied to the real normed space Y , with the newly obtained local compactness; it requires no separately assumed completeness of Y .

Main citations

- Exact declaration and its source — `MathlibAnnex.Sphere.finiteDimensional_codomain`
- The radial homeomorphism — `MathlibAnnex.Sphere.radialExtensionHomeomorph`
- Finite-dimensionality in the opposite direction — `MathlibAnnex.Sphere.finiteDimensional_domain`
- Finite-dimensional real normed spaces are proper — `FiniteDimensional.proper_real`
- Local compactness is preserved by a homeomorphism — `Homeomorph.locallyCompactSpace_iff`
- Local compactness implies finite-dimensionality — `FiniteDimensional.of_locallyCompactSpace`

Lean source signature (exact)

```
theorem finiteDimensional_codomain
  [FiniteDimensional ℝ X]
  (e : Metric.sphere (0 : X) 1 ≈i Metric.sphere (0 : Y) 1) :
  FiniteDimensional ℝ Y
```

In the source	Mathematical meaning
(e : Metric.sphere (0 : X) 1 ≈ _i Metric.sphere (0 : Y) 1)	The same bijective chord-distance isometry $\Delta : S_X \rightarrow S_Y$, where $S_X = \{x : \ x\ _X = 1\}$ and $S_Y = \{y : \ y\ _Y = 1\}$. The surrounding binders make X, Y real normed spaces.
[FiniteDimensional ℝ X]	The input hypothesis $\dim_{\mathbb{R}} X < \infty$. Only X is initially assumed finite-dimensional.
FiniteDimensional ℝ Y	The conclusion $\dim_{\mathbb{R}} Y < \infty$. Local compactness of Y is derived in the proof through the radial homeomorphism, not an extra hypothesis.

▼ **Exact surrounding binder context (separate excerpt)**

```
namespace MathlibAnnex
namespace Sphere

universe u v

variable {X : Type u} {Y : Type v}
  [NormedAddCommGroup X] [NormedSpace ℝ X]
  [NormedAddCommGroup Y] [NormedSpace ℝ Y]
```

▼ **Exact content identity**

Declaration: `MathlibAnnex.Sphere.finiteDimensional_codomain`

Accepted content SHA-256: `6c804f309c6c2063cd8517656164e90d1419c13749b356e1e11001c2a2747929`

Accepted source guide SHA-256: `1e6ec3108116a9319e90c42787a35a1bca0dda4996e96387e1a45a05b56b6902`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`